

1.7 DFE Case Study: Transportation Industry

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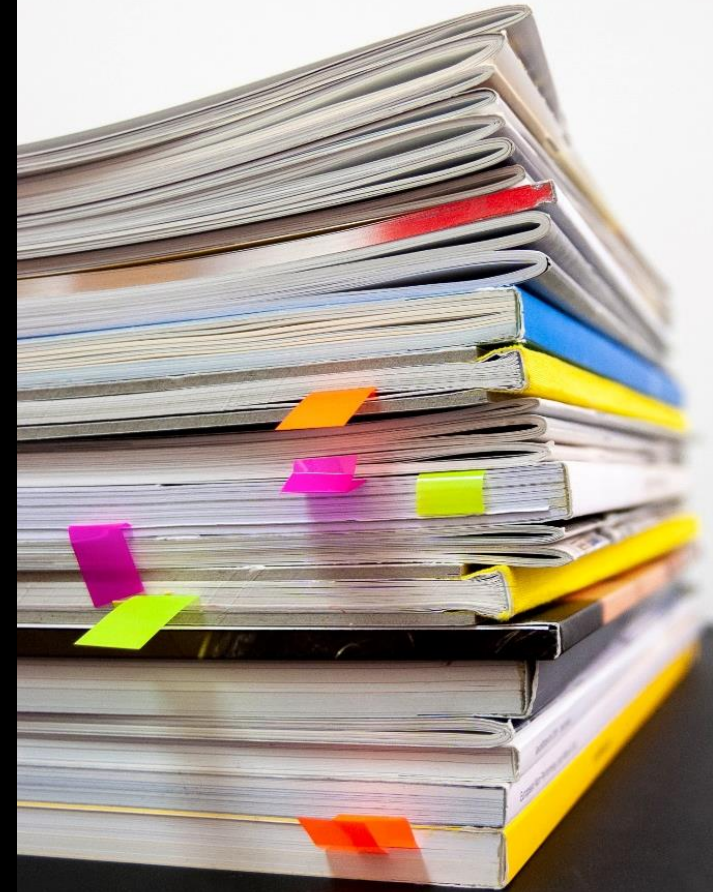


What this lecture covers...

- DFE report advice
- A look at the transportation both overseas and NZ
- DFE Case studies: General Motors and DuPont

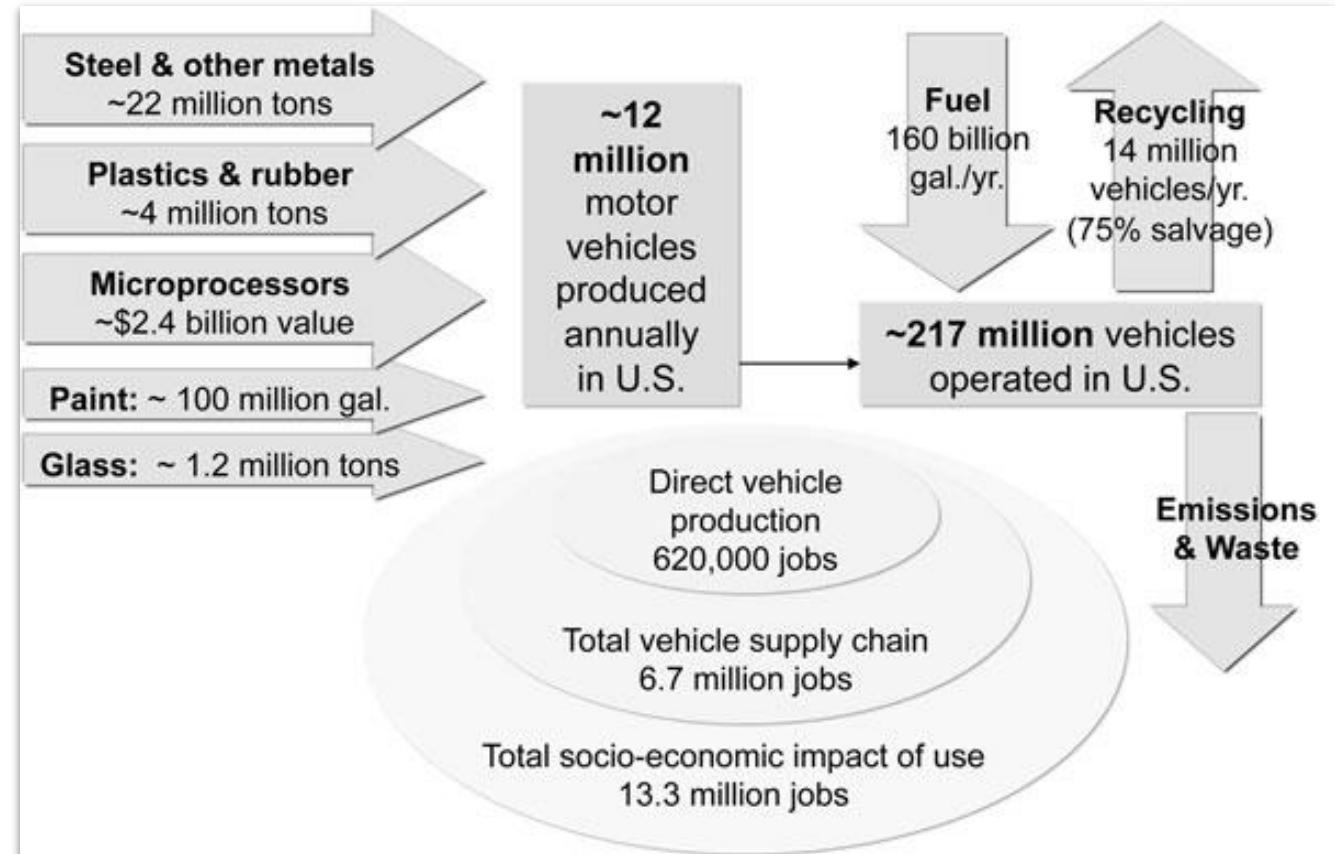
DFE Report

- Each student will take responsibility for one product / process, identified in the project brief. The analysis of each product / process should include the following subsections:
 - *Systems and Life Cycle thinking*
 - *Design for Dematerialization*
 - *Design for Detoxification*
 - *Design for Revalorization*
 - *Design for Capital Protection and Renewal*
- Consider different metrics for each “Design for *” subsection



Transportation Industry

- Address the grand societal demand for human mobility
- Transportation systems, and the automotive industry in particular, have an enormous impact on both the economy and the environment.

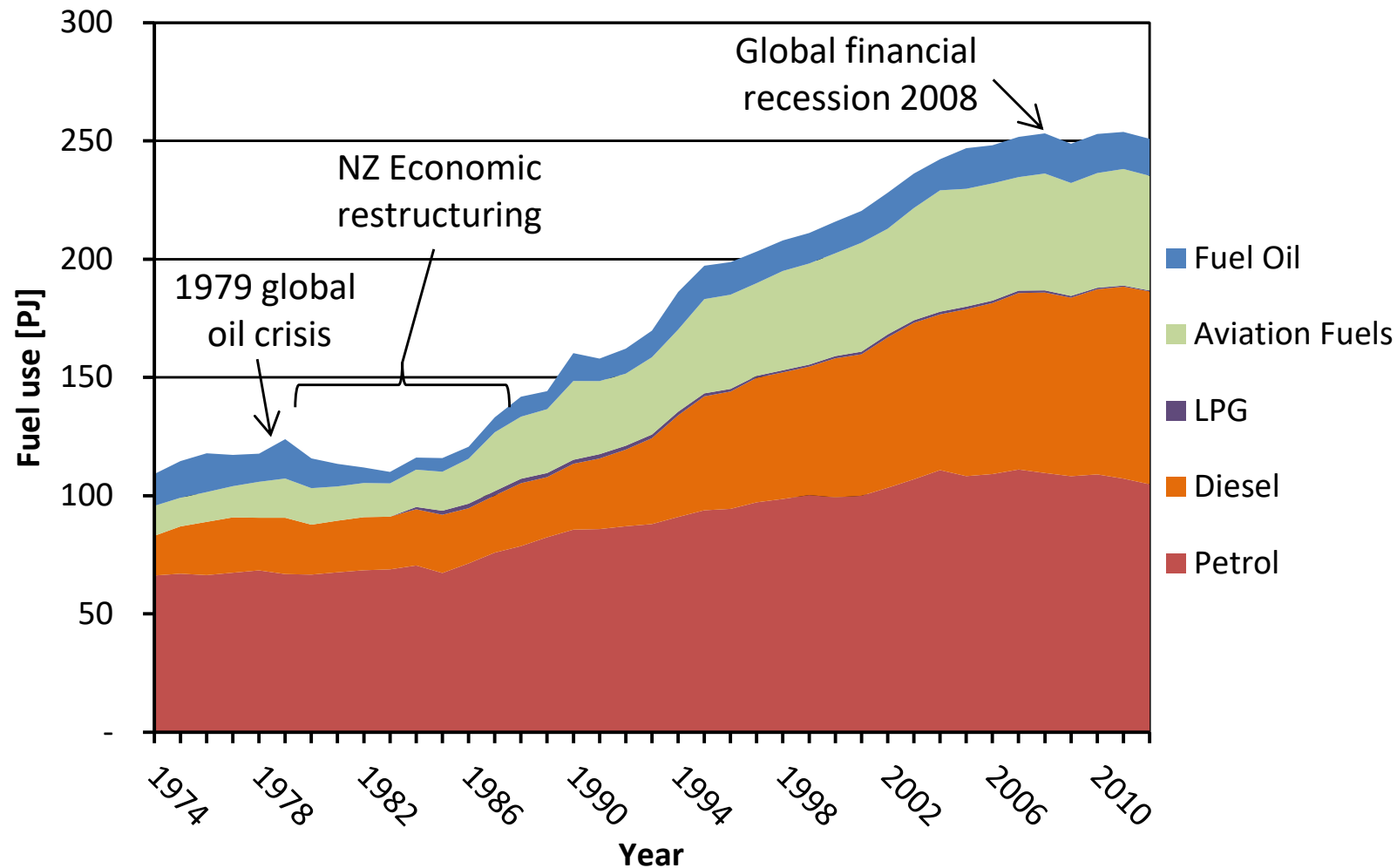


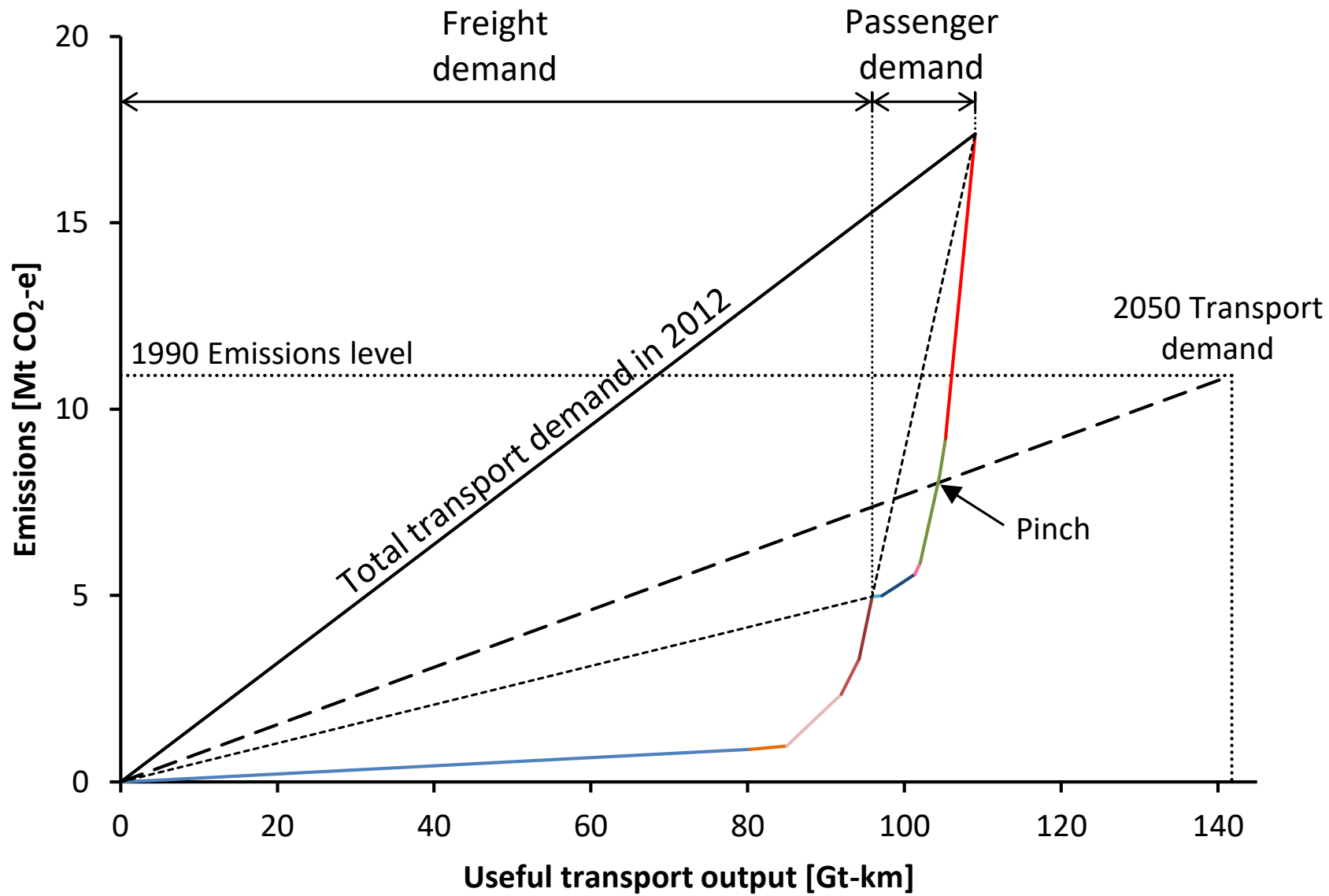
Environmental impacts

- The continued growth in mobility demand has led to increasing concerns about potential impacts on the environment
 - **Examples:** Atmospheric pollution and greenhouse gas emissions, accumulation of solid wastes (such as tyres and scrap plastics from obsolete vehicles).
 - Other adverse impacts include noise, traffic congestion, and highway fatalities.



New Zealand Transport Energy Use





EMISSIONS VS TRANSPORTATION

- | | | | |
|-------------------|----------------|-------------------|-----------------|
| — Marine cargo | — Freight Rail | — Truck (>15 t) | — Truck (<15 t) |
| — LCV | — Ferry | — Cruises | — Bus |
| — Inter-City Rail | — Air travel | — Intra-City Rail | — LPV & MC |

Many Options

- **Modes** of transport: air, sea, rail, and highway
- Different **fuel** sources: fossil fuels, biofuels, electricity, and hydrogen
- Different **infrastructure** configurations
- Different **settings**: developing and developed nations, urban and rural, and across different geographic and climatic settings
- Different active ongoing research in sustainable mobility:
 - **1. Technological innovation**, including alternative materials, vehicle designs, energy sources, propulsion systems, and transportation networks that are safer, more effective, and more environmentally benign.
 - **2. Technology assessment** to determine the feasibility, eco-efficiency, sustainability, and resilience of alternative mobility technology combinations

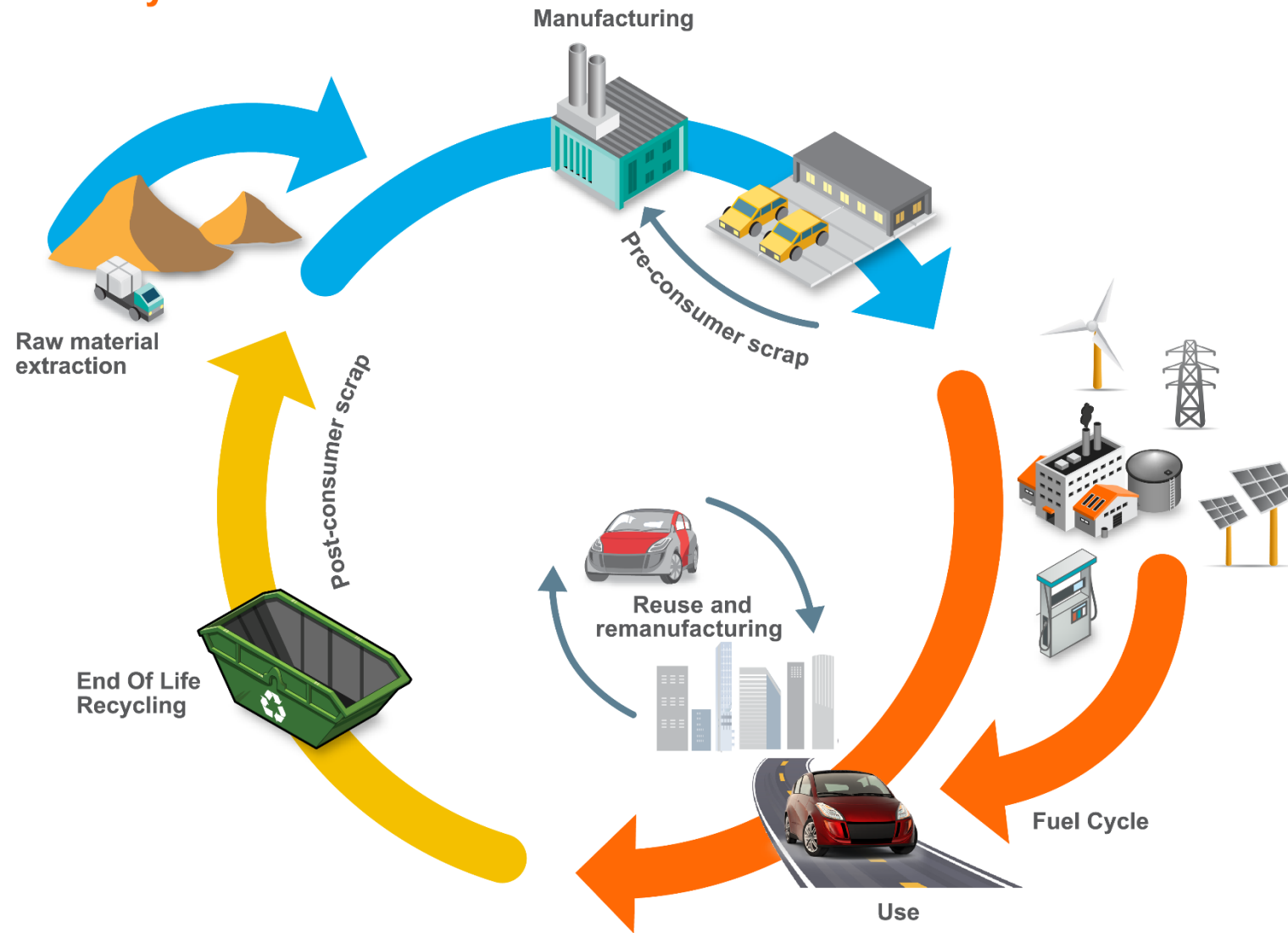


Life Cycle Thinking

Each mode of transport involves a variety of different systems with different life-cycle considerations

Transport Mode	Vehicle Systems	Energy Systems	Operational Systems	Infrastructure Systems	Life-Cycle Systems
Air	Aircraft, Dirigibles	Liquid Fuels, Gases	Air Traffic Control	Airports, Provisioning	Maintenance, Recycling
Water	Watercraft, Amphibious	Liquid Fuels, Wind Power	Harbor Control	Ocean Terminals, Provisioning	Maintenance, Recycling
Rail	Conventional, High-Speed	Liquid Fuels, Electric	Monitoring, Dispatching	Railways, Provisioning	Maintenance, Recycling
Road	Automobiles, Motorcycles, Moto-Rickshaws Trucks, etc.	Liquid Fuels, Fuel Cells, Electric, Hybrid, etc.	Monitoring, Congestion Management	Roadways, Bridges, Refueling Stations	Maintenance (Vehicle/Road), Reverse Logistics
Non-Powered	Pedestrian, Bicycles, Animal-Drawn	N.A.	N.A.	Roads, Dedicated Pathways	Maintenance, Animal Husbandry

Vehicle life cycle



Life Cycle thinking

- Impacts occur at many points along the value chain
- Energy use and greenhouse gas emissions are dominated by the vehicle use stage
- Largest contributions to other wastes occur in the materials acquisition stage
- One of the most direct strategies for reducing energy consumption during vehicle use is the use of lighter-weight materials
 - *Example: Each kilogram of aluminium replacing mild steel, cast iron, or high strength steel saves between 13 and 20 kilograms of greenhouse gas emissions depending on the component*

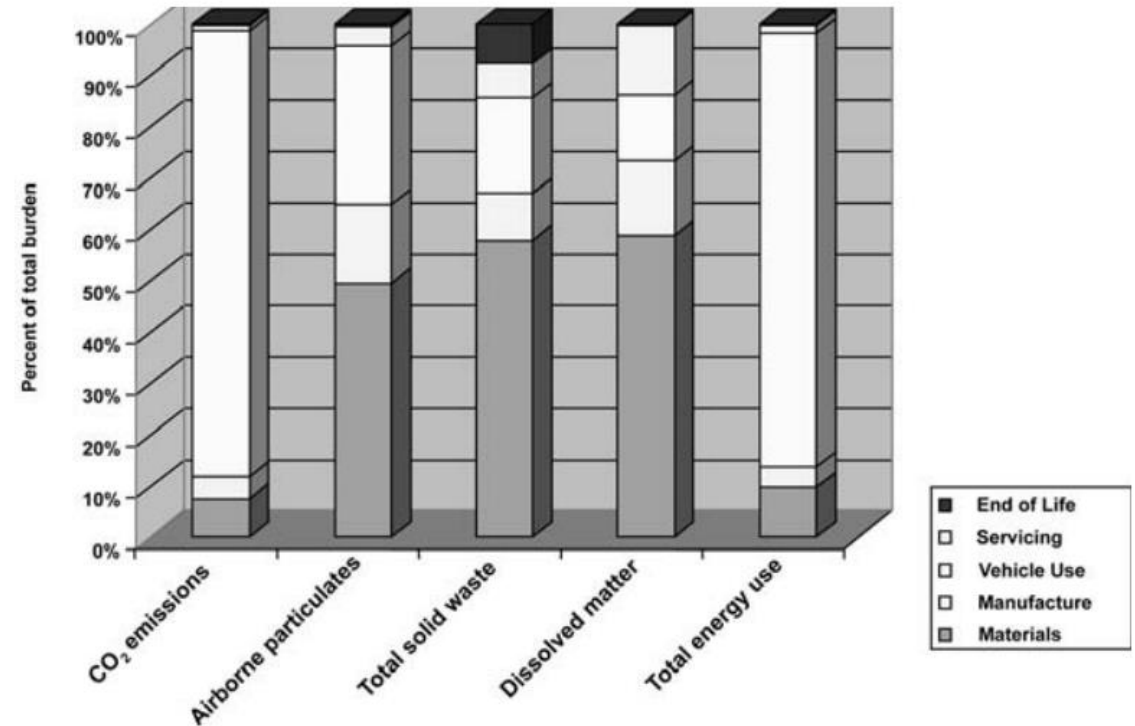
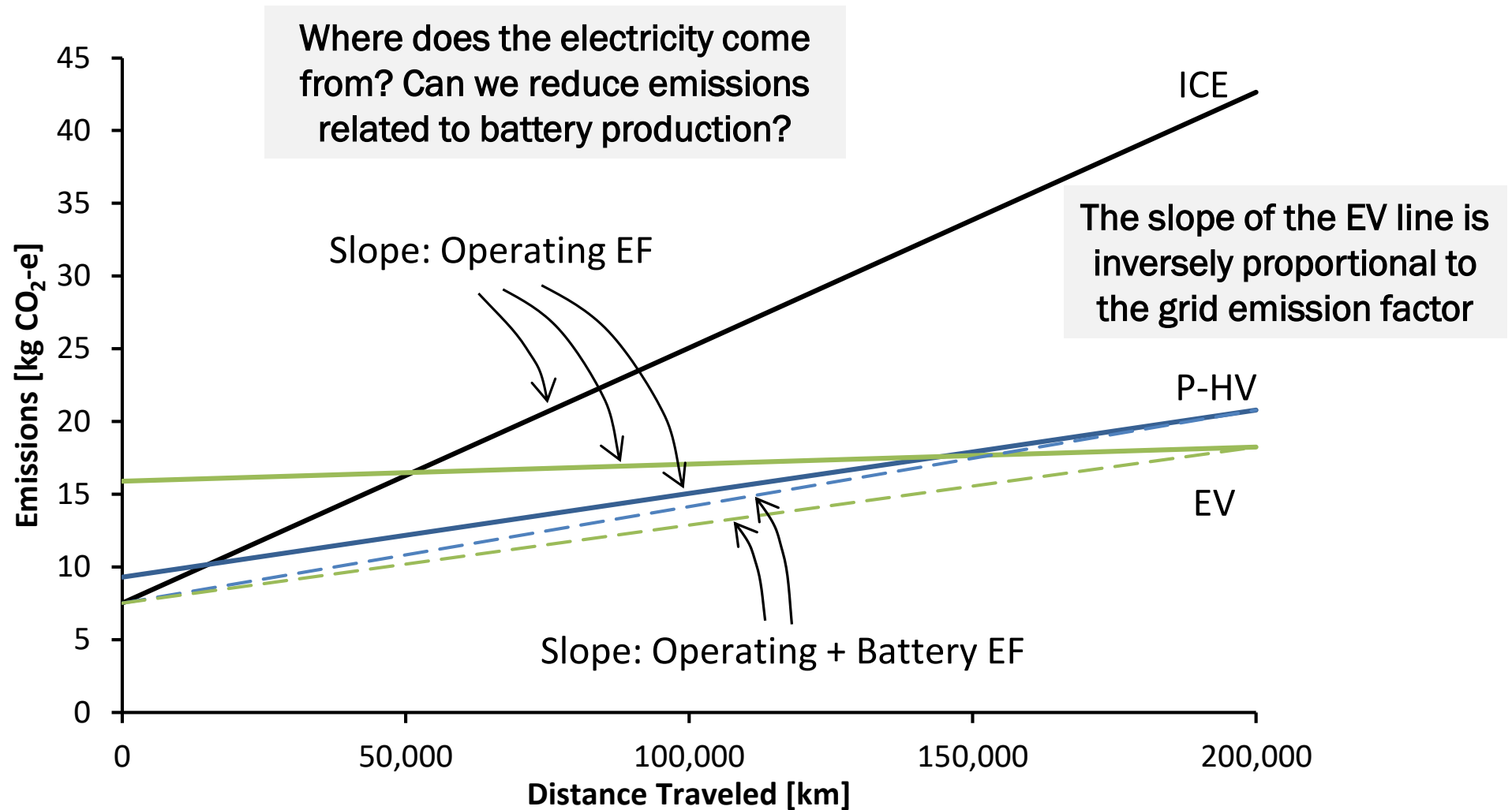


FIGURE 12.2 Life-cycle environmental burdens due to production and use of motor vehicles [4].

Electric vs Internal Combustion Engine for NZ



DFE initiative examples

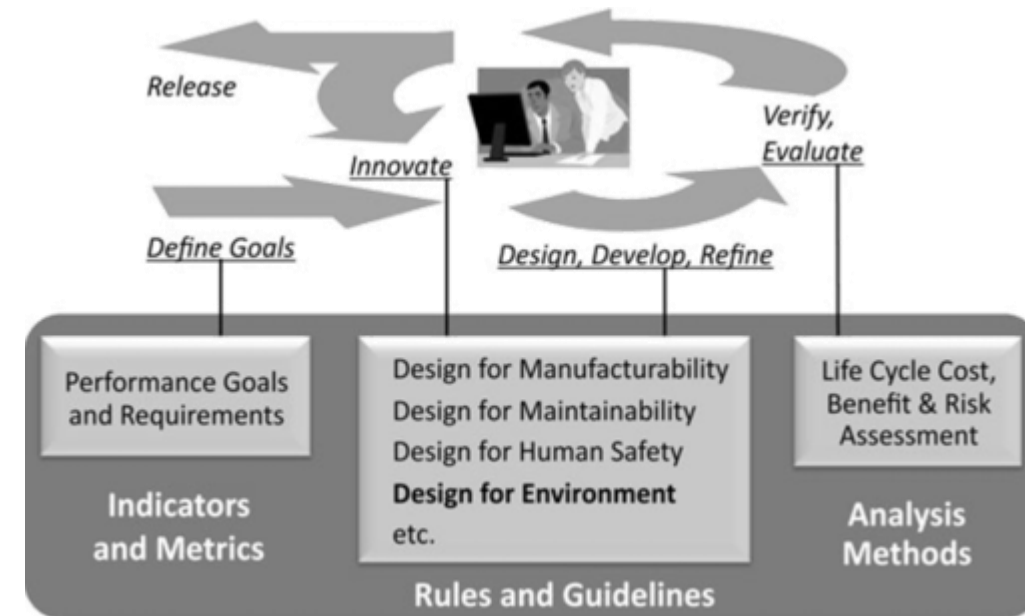
- Address energy and environmental goals at various points in the life cycle:
 - **NetJets** (fractional jet ownership company): sponsors the Next Generation Jet Fuel Project at Princeton U. (producing jet fuel with near-zero green house gas emissions)
 - **Car sharing companies (Mevo, Zipcar)**: reduces fuel consumption and emissions. Reduces strain on the urban parking infrastructure
 - **Ford Motors**: flexible, polyurethane foams that replace traditional, petroleum-derived polyols reduced CO2 emissions from raw materials/ manufacturing



GENERAL MOTORS

Case study: General motors

- General Motors (GM), founded in 1908, is one of the world's largest automakers
- Established environmental policies and principles in the early 1990s
- reports its progress with respect to global environmental indicators/metrics:
 - *energy use*
 - *water use*
 - *greenhouse gas emissions*
 - *recycled and nonrecycled waste*



1. GM's Advanced Propulsion Technology Strategy

- Recognizing that alternative sources of propulsion will be essential, GM focused on displacing petroleum through energy efficiency and energy diversity:
 - *Guidelines: GM's Advanced Propulsion Technology Strategy (GMAPTS)*





Objectives

- Establish sustainability through diversified sources of energy
- Displace petroleum in order to alleviate the mismatch between supply and demand and reduce U.S. dependence on foreign imports subject to uncontrollable risks (i.e., geopolitical, oil refinery capacity, natural disasters, wars, etc.)
- Reduce GHG emissions.

Strategies & Techniques



- Continue to improve conventional powertrains,
 - *recognizing that gas engines, diesel engines, and transmissions will be the principal propulsion systems for the foreseeable future.*
- Develop alternative fuels,
 - *including biofuels blends such as E85 ethanol and B20 biodiesel as well as synthetic fuels derived from coal, natural gas, and/or biomass.*
- Increased electrification of the automobile through expansion of the hybrid portfolio,
 - *including plug-in hybrids, EV range extenders, and ultimately, hydrogen fuel cells.*

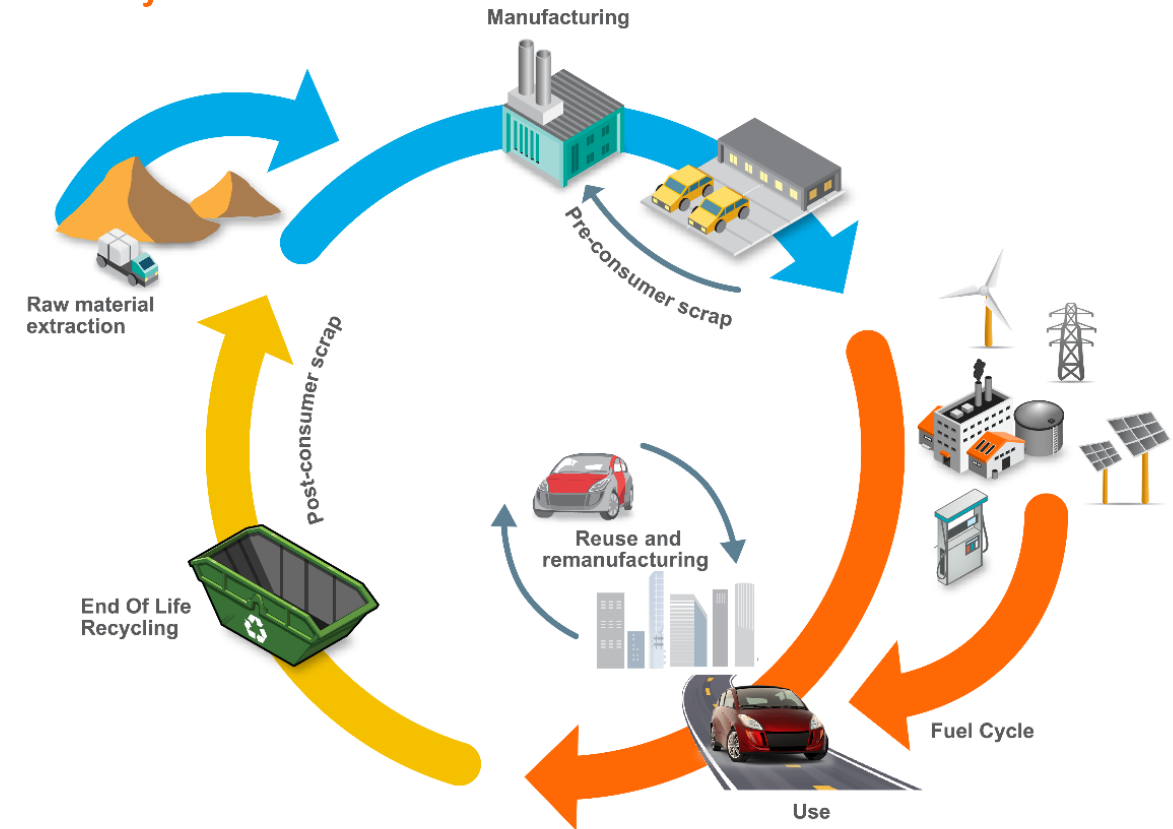
Outcome: Chevrolet Volt (2007)

- Powered by GM's Voltec electric propulsion system
- Uses electricity as its primary energy source and gasoline as its secondary energy source for propelling the vehicle.
- "We believe this is the biggest step yet in our industry's move away from our historic, virtually complete reliance on petroleum to power vehicles."
- More than 75% of drivers in the US commute <65 km/day. For these drivers, a fully charged Chevy Volt will use no gasoline and produce no tailpipe emissions.
- Added benefit (synergy): Use of electricity over gasoline results in a reduction in running costs of about \$1,200 per year for an average driver (US).

2. GM's Global By-Products Management

- Another set of guidelines that focuses on GM's production and supply chain processes.
- Aimed at establishing a single global process to leverage all manufacturing by-product materials, maximize values, and reduce the environmental footprint of GM.
- Objective: No more landfilling

Vehicle life cycle



Strategies and Techniques



- Waste aluminium generated at GM facilities is sent to GM foundries to be reused to produce engine and transmission components.
- Steel, alloy metals, and paper are sent to recyclers to be made into a variety of products.
- Used oil is reconditioned for reuse in GM facilities.
- Wood pallets are reused, rebuilt ground into landscape chips, or sent to waste-to-energy facilities.
- Empty drums, totes, and containers are refurbished and reused again and again.
- Cardboard is collected, compacted, and sold for making new cardboard materials.

Outcome

- Resulted in development of landfill-free plants: over 96% of waste materials are recycled or reused, and about 3% is converted to energy at waste-to-energy facilities
- Some of the materials recycled at GM's zero landfill sites include 630,000 tons of scrap metal, 8,000 tons of wood, 7,500 tons of cardboard and 1,200 tons of plastic.
- 3.65 million metric tons of carbon dioxide emissions were avoided. Additionally,
- Using recycled by-products to make new products also reduced manufacturing costs, compared to using virgin materials (synergy).



DUPONT

DuPont: Eco-Efficient Automotive Paint

- DuPont is one of the world's leading chemical companies
- incorporated sustainability principles into its product development efforts
- **Issue:** Traditional coating processes use significant amounts of solvent and energy to apply paint to automotive vehicles, and the resulting volatile organic compound (VOC) emissions are regulated toxic substances
- **Solution:** DuPont introduced a breakthrough system called EcoConcept.

EcoConcept

- Is a primerless, water-based coating system: This eliminates the need one complete coating layer that is usually a solvent-borne product.
- Combines both product and process innovations to reduce production costs and energy usage, increase speed and productivity, and lower VOC emissions



EcoConcept Outcome:

- Process energy and **solvent emissions** are reduced by approximately 25%.
- **Material consumption** is lowered by approximately 20–25%.
- **Greenhouse and acidification** gases are reduced by approximately 30%.
- Paint shop total **time and costs** are reduced by approximately 20–25%.
- The overall supply chain **footprint** of automotive paint acquisition is reduced by about 20% due to lower mass throughput

1.8 DFE Case Study: Food and Beverage Industries

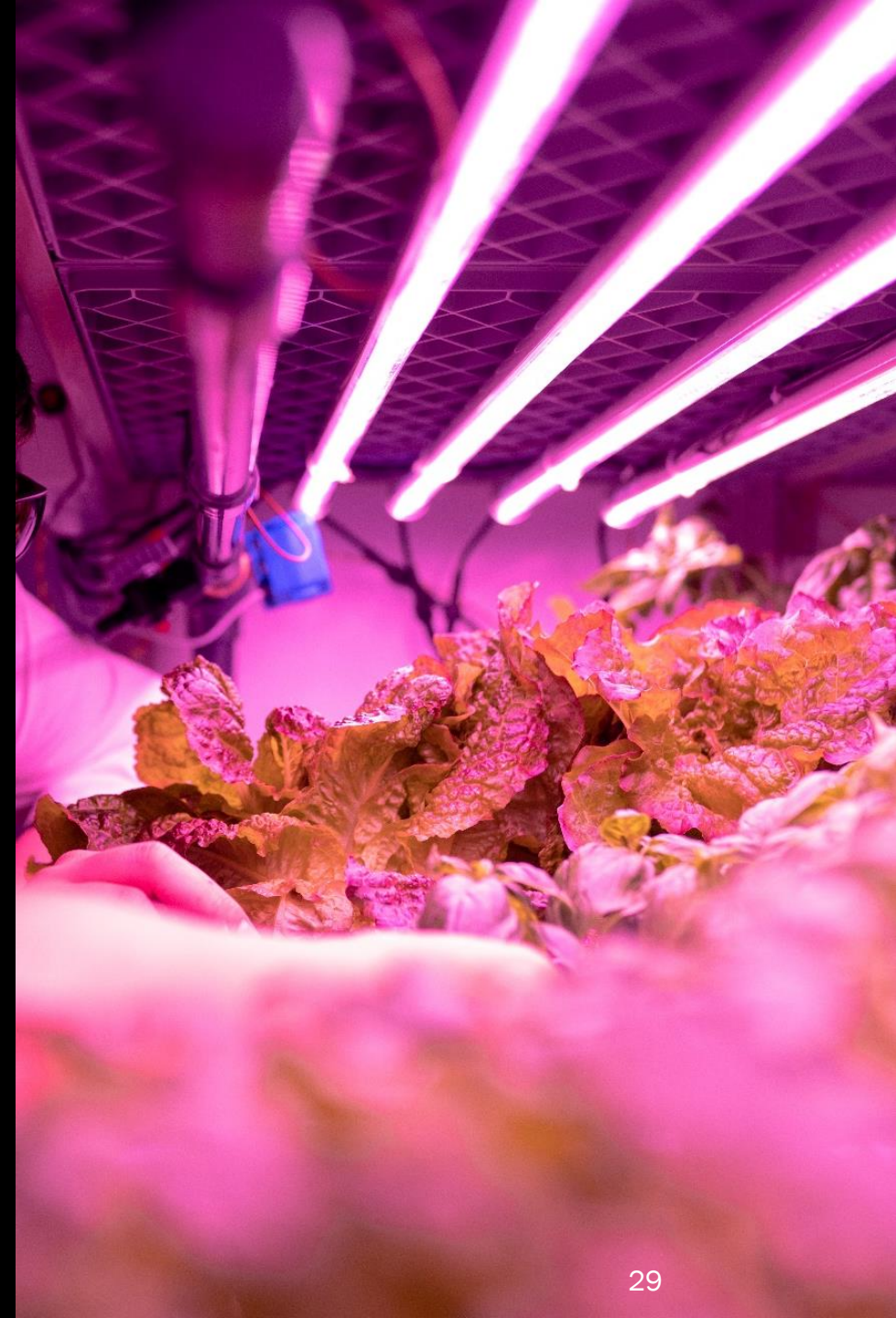
Dr Peter Kovalsky

Food and Beverage Industries

- The agricultural sector accounts for over 50% of habitable land around the world, 70% of water use, the highest use of chemicals, and from 25% to 40% of all greenhouse gas emissions (WWF study)
- Meanwhile, hundreds of millions of people live in poverty, own no land, and are chronically hungry
- Global food demand will double in 50 years

Environmental impacts are complex

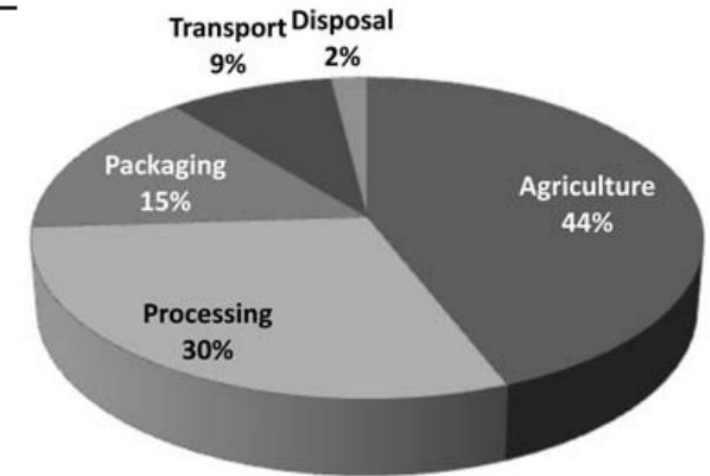
- Health consciousness and rising consumer interest in “natural” and “organic” foods
- Safety/contaminations concerns e.g. E.coli or melamine contamination in imported commodities
- Genetically engineered crops controversy



Examples of progressive industry practices:

- **Kraft Foods:** conversion of waste biomass, such as whey and meat, to energy. Resulted in decrease in absolute energy use by 14% from 2001 to 2006,
- **Heinz:** developed tomato seeds using traditional breeding techniques that remain ripe longer in the field, are more disease-resistant so they require less pesticide use, and produce a higher yield, thus reducing consumption of water and fertilizer
- **PepsiCo:** worked with the Carbon Trust in the United Kingdom to affix a carbon footprint label to its Walker's Crisps

FIGURE 15.1 Proportion of carbon emissions for Walker's Crisps from each life-cycle stage.







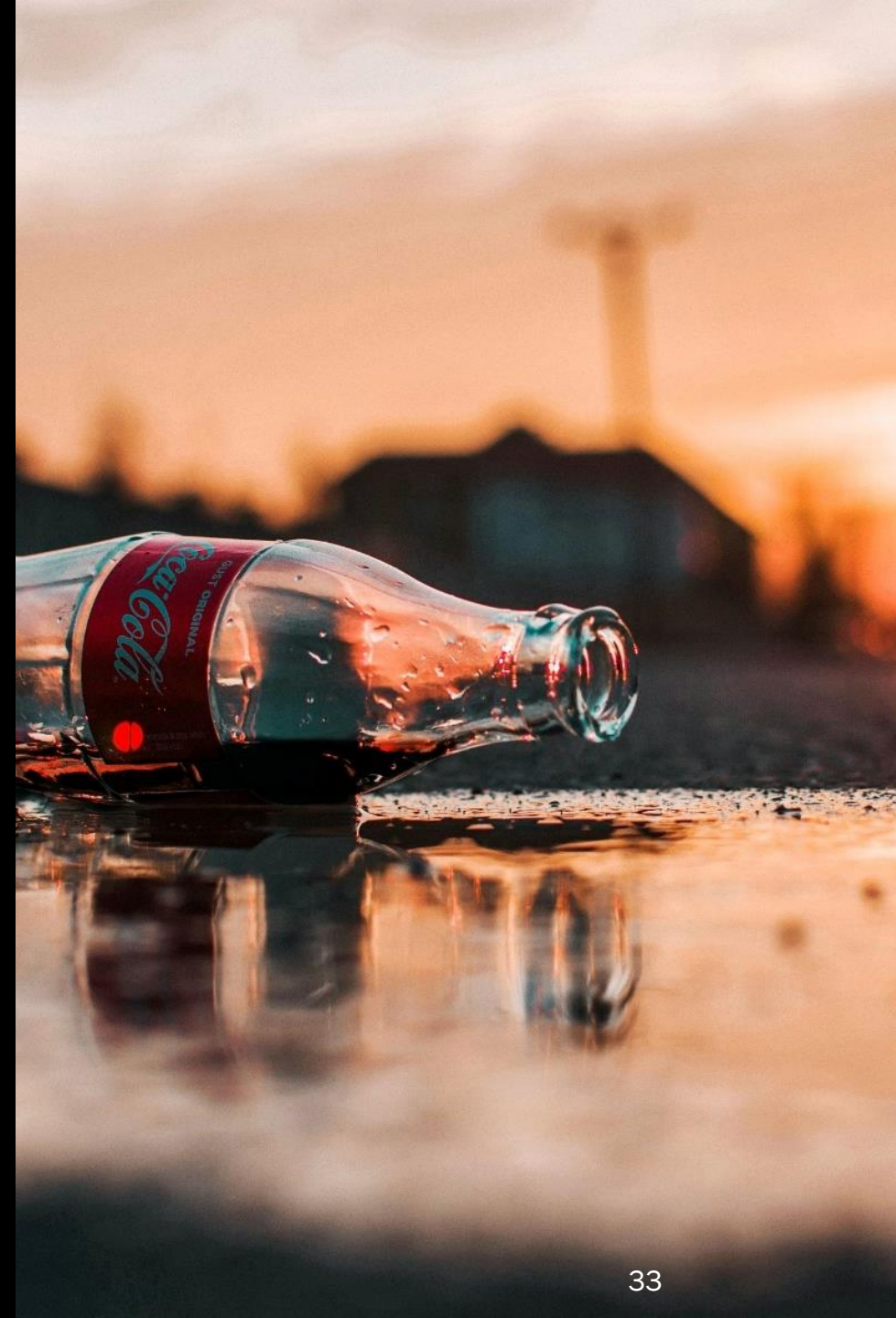
Coca-Cola

- The world's largest beverage company
- The system's environmental impacts are primarily due to plant operations and distribution networks, as well as vending machines and coolers
- Key environmental **indicators**:
 1. *Water stewardship*
 2. *Sustainable packaging*
 3. *Energy and climate protection*

Metrics

Coke concentrates on three principal areas of environmental responsibility, each with explicit performance metrics

1. **Water stewardship:** primarily measured by the water use ratio, i.e., litres of water per litre of product.
2. **Sustainable packaging:** measured in multiple ways including % recycle and reuse of PET (polyethylene terephthalate) and aluminium packaging.
3. **Energy and climate protection:** primarily measured by the energy use efficiency ratio, i.e., megajoules per litre of product.



COMPOST

Sustainable Packaging

DfE efforts include:

- Designing packages that use less material without sacrificing quality
- Investing in technologies and recovery systems that enable greater use of recycled materials

NO PLASTIC

WASTE

ALL PLASTIC BAGS

RECYCLE

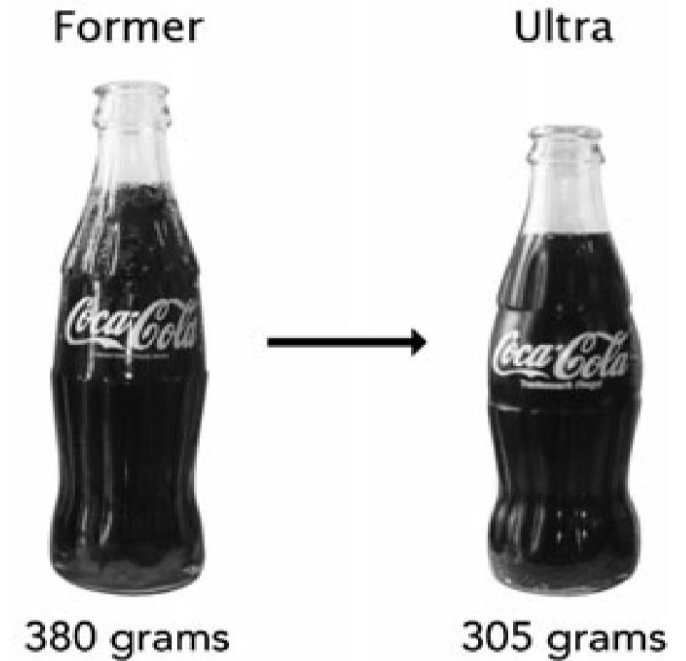


NO CUPS

Sustainable Packaging outcomes:

Designing packages that use less material without sacrificing quality:

- Reduced the weight and improved the impact resistance of their glass bottles.
- New bottles are 40% stronger, 20% lighter and 10% less expensive than traditional bottles.
- Saved 89,000 metric tons of glass in 2006, the CO2 equivalent of planting more than 13,000 acres of trees.



Sustainable Packaging outcomes:

Investing in technologies and recovery systems that enable greater use of recycled materials:

- Invested more than \$60 million to build the world's largest plastic-bottle-to-bottle recycling plant in Spartanburg, S.C.
- The plant produces approximately 100 million pounds of food-grade recycled PET (polyethylene terephthalate) plastic for reuse each year
- Over the next ten years it is expected to eliminate the production of one million metric tons of carbon dioxide emissions.
- Recycling plastic for reuse yields financial benefits, requires less energy than producing bottles with virgin materials, and reduces waste and greenhouse gases.

Energy and climate protection: Sustainable Refrigeration

- Refrigeration equipment is the largest contributor to Coke's climate footprint
- Sustainable refrigeration program objectives:
 - *improve overall energy efficiency,*
 - *reduce greenhouse gas (GHG) emissions like hydrofluorocarbons (HFCs).*



Sustainable Refrigeration outcomes:

1

Transitioning to HFC-free insulation foam for new equipment:

- *Generates 75% fewer direct GHG emissions than traditional equipment*

2

Deploying HFC-free equipment using CO2 as the refrigerant:

- *1,300 times less potent GHG emissions than HFCs.*

3

Developing a proprietary energy management system (EMS) for coolers and vending machines:

- *Delivers energy savings of up to 35%*



GENERAL MILLS

General Mills: Nourishing the Future

- In addition to its popular cereal brands, such as Cheerios, General Mills brands include Betty Crocker, Pillsbury, Yoplait, and Haagen-Dazs ice cream.
- Its sustainability program around the theme of nourishment:
 1. *Nourishing lives by providing food products that meet or exceed expectations for safety, quality, convenience, value etc*
 2. *Nourishing communities by volunteering, donating money and food, supporting minority-owned businesses, and creating healthy and safe workplaces for employees.*
 3. *Nourishing the future by establishing sustainable business and manufacturing practices that minimize the company's environmental footprint*



Environmental Goals and Indicators

- Set five-year goals for reducing its key environmental indicators across its many plants:
 - *Reduce water usage rate by 5%*
 - *Reduce energy consumption rate by 15%*
 - *Reduce greenhouse gas emission rate by 15%*
 - *Reduce solid waste generation rate by 15%*





Outcomes

- Missouri plant reduced water usage by more than 14 million gallons per year
 - *Reusing water from air compressors for its cooling towers*
 - *Switching to more dry cleaning than wet cleaning*
 - *Installing a high pressure, low volume cleaning system.*
- Ohio plant reduced landfilled waste by about 42%
 - *Selling waste as recyclable material; for example, the plant has found buyers for its corrugated cardboard, mixed paper, film, and other trash. Even excess food discarded in the production process is being sent to farms to feed pigs and chickens.*